

Big Multiplication

Every graded set, every week — 4 weeks, 20 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Multiples of Ten

Get these automatic and the algorithm gets easy.

Warm-Up 11 ITEMS • RECALL

1 20×30

2 40×50

3 60×70

4 30×300

5 80×20

6 90×40

7 30×20

8 50×40

9 60×60

10 70×70

11 90×30

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. 400×60

13. 700×800

14. 25×40

15. 120×50

16. 36×200

17. 300×40

18. 500×600

19. 150×40

20. 220×30

21. 45×200

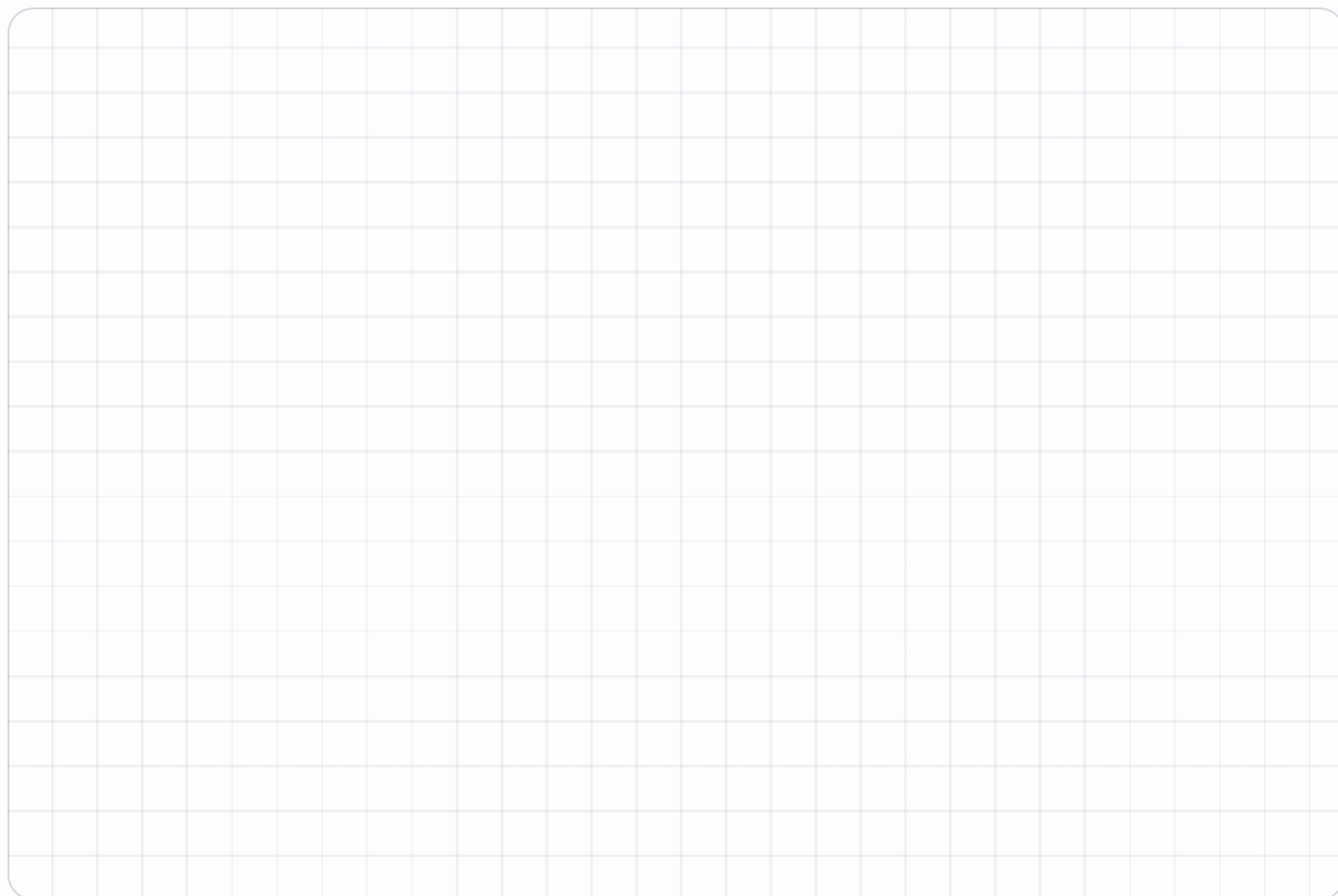
★ Challenge NOT GRADED • REASON IT OUT

22. 250×400
.....

23. 1250×80
.....

24. 1500×60
.....

25. 3500×200
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Partial Products

Four boxes, four numbers, one total.

Warm-Up 12 ITEMS • RECALL

1 23×10

2 23×4

3 Add 230 and 92

4 So 23×14 is

5 31×20

6 31×2

7 20×4

8 3×7

9 20×7

10 30×6

11 5×8

12 40×3

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 31×22

14. 45×23

15. 64×35

16. 128×12

17. 216×30

18. 24×6 as 20×6 plus 4×6

19. 32×7

20. 46×5

21. 58×4

22. 27×8

The Standard Algorithm

Stack it, but keep an estimate in your head.

Warm-Up 12 ITEMS • RECALL

1 14×12

2 25×11

3 32×21

4 18×15

5 43×13

6 52×12

7 12×11

8 20×15

9 11×11

10 13×10

11 15×4

12 25×4

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. 237×45

14. 418×27

15. 506×34

16. 725×18

17. 1234×12

18. 34×27

19. 52×19

20. 46×38

21. 63×24

22. 75×32

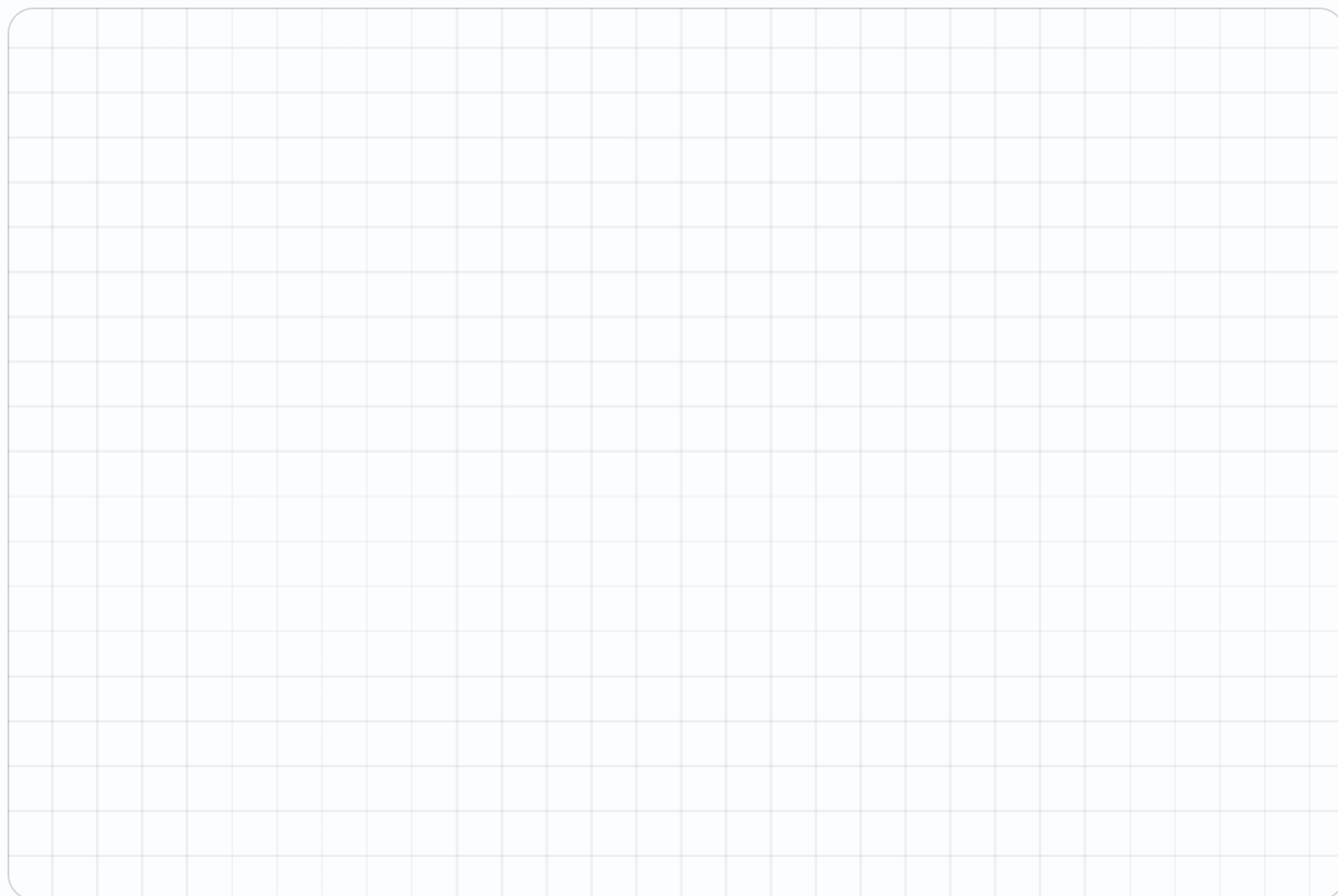
★ Challenge NOT GRADED • REASON IT OUT

23. 2450×36
.....

24. 3081×47
.....

25. 148×36
.....

26. 267×54
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Estimate to Check

Round both numbers, multiply, compare. A wrong answer should look wrong.

Warm-Up 12 ITEMS • RECALL

1 Estimate 19×21 by rounding both

2 Estimate 48×52

3 Estimate 31×29

4 Estimate 39×11

5 Estimate 62×18

6 Estimate 78×22

7 Estimate 21×19

8 Estimate 49×11

9 Estimate 32×28

10 Estimate 58×22

11 Estimate 81×19

12 Estimate 68×31

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Estimate 412×19

.....

14. Estimate 289×31

.....

15. True value of 412×19

.....

16. Estimate $5,120 \times 48$

.....

17. Estimate 197×203

.....

18. Estimate 312×21

.....

19. True value of 312×21

.....

20. Estimate 489×29

.....

21. True value of 489×29

.....

22. Estimate 198×402

.....



Working space



Error journal

Poker & Floor Plan

Enrichment. Area two ways, and they must agree.

NAME _____

DATE _____

RIGHT ____ OF ____ TRIED

✦ Warm-Up

4 ITEMS • RECALL

1 Area of a 12 by 10 room, in squares

2 Area of a 20 by 15 room

3 40×50

4 12×12

◆ Core

GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. A 14 by 12 room scaled by 3: new length

6. That scaled room's area (42 by 36)

7. 36×24

8. 143×22

★ Challenge NOT GRADED • REASON IT OUT

9. Digits 3, 4, 7, 2 — biggest product from two 2-digit numbers: type it (73 × 42)

10. Area of a 26 by 34 room

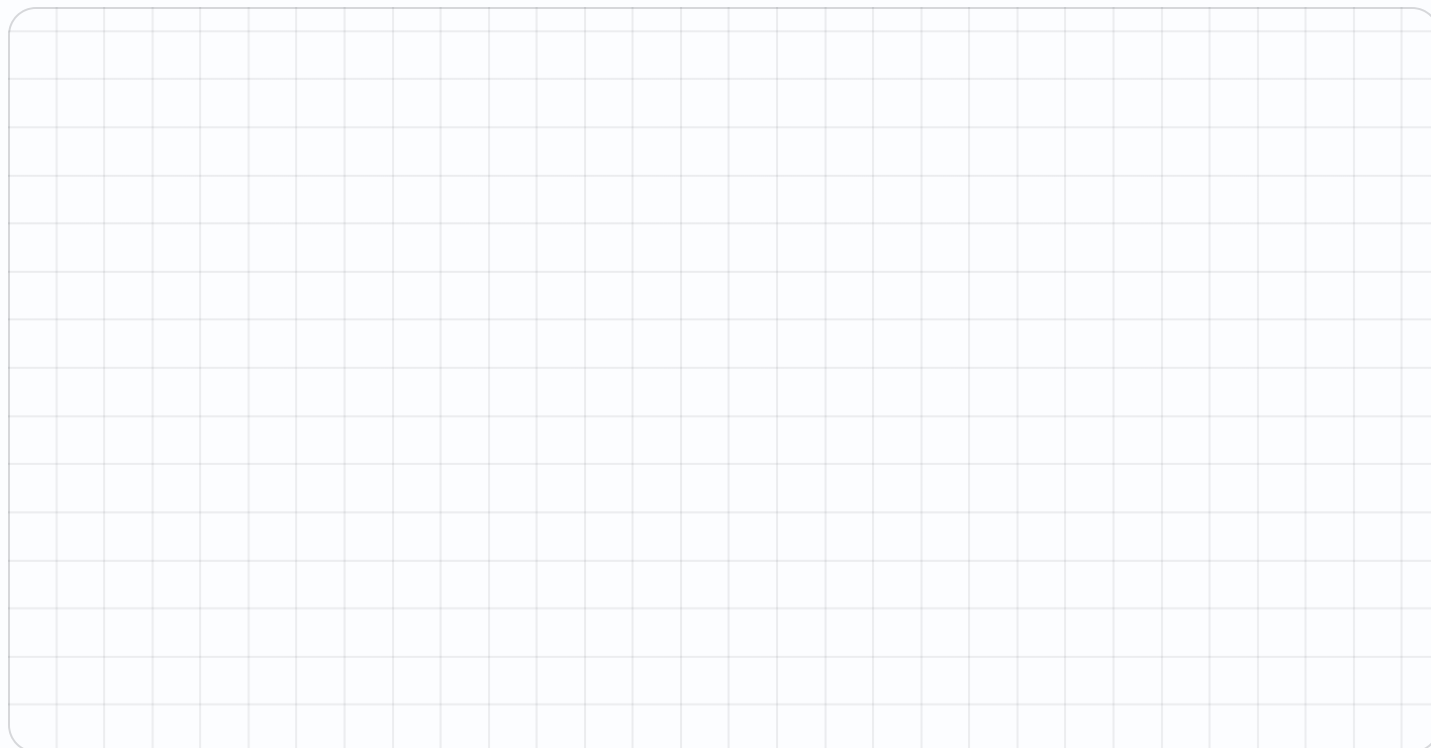
11. Double both sides of a 15 by 20 room. Area multiplies by

12. 1250×48

13. A 24 by 36 room — square feet

14. Doubling both sides of that room — the new area

Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE



! Error journal ONE SENTENCE PER MISTAKE

Mark the reasoning, not the handwriting.

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

1.1 • Multiples of Ten

OUT OF 21

Warm-Up • 1-11 • 600 | 2000 | 4200 | 9000 | 1600 | 3600 | 600 | 2000 | 3600 | 4900 | 2700

Core • 12-21 • 24000 | 560000 | 1000 | 6000 | 7200 | 12000 | 300000 | 6000 | 6600 | 9000

Challenge • 22-25 • 100000 | 100000 | 90000 | 700000

1.2 • Partial Products

OUT OF 22

Warm-Up • 1-12 • 230 | 92 | 322 | 322 | 620 | 62 | 80 | 21 | 140 | 180 | 40 | 120

Core • 13-22 • 682 | 1035 | 2240 | 1536 | 6480 | 144 | 224 | 230 | 232 | 216

Challenge • 23-26 • 7344 | 9022 | 744 | 1652

1.3 • The Standard Algorithm

OUT OF 22

Warm-Up • 1-12 • 168 | 275 | 672 | 270 | 559 | 624 | 132 | 300 | 121 | 130 | 60 | 100

Core • 13-22 • 10665 | 11286 | 17204 | 13050 | 14808 | 918 | 988 | 1748 | 1512 | 2400

Challenge • 23-26 • 88200 | 144807 | 5328 | 14418

1.4 • Estimate to Check

OUT OF 22

Warm-Up • 1-12 • 400 | 2500 | 900 | 400 | 1200 | 1600 | 400 | 500 | 900 | 1200 | 1600 | 2100

Core • 13-22 • 8000 | 9000 | 7828 | 250000 | 40000 | 6000 | 6552 | 15000 | 14181 | 80000

Challenge • 23-26 • 884 | 10000 | 200000 | 1472

Fri • Poker & Floor Plan

OUT OF 8

Warm-Up • 1-4 • 120 | 300 | 2000 | 144

Core • 5-8 • 42 | 1512 | 864 | 3146

Challenge • 9-14 • 3298 | 884 | 4 | 60000 | 864 | 3456

Multiples of Ten First

40 × 60 before 43 × 67. Every time.

Warm-Up 12 ITEMS • RECALL

1 20 × 30

2 40 × 60

3 50 × 50

4 70 × 30

5 80 × 40

6 90 × 90

7 30 × 30

8 20 × 50

9 60 × 40

10 70 × 80

11 90 × 20

12 50 × 60

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. 400 × 70

14. 600 × 500

15. 250 × 40

16. 120 × 60

17. 35 × 200

18. 600 × 40

19. 700 × 300

20. 350 × 20

21. 140 × 50

22. 25 × 400

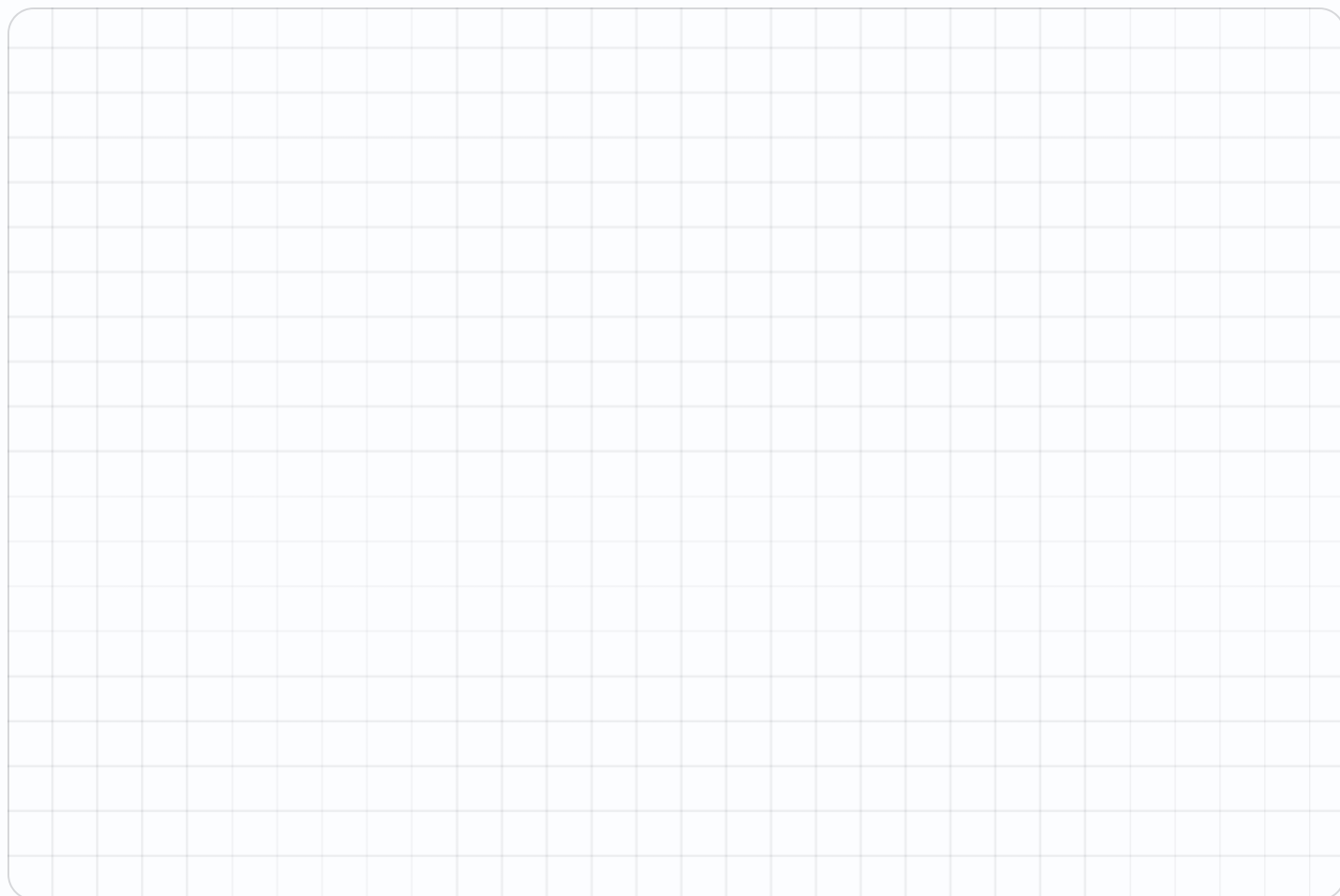
★ Challenge NOT GRADED • REASON IT OUT

23. 1250×80
.....

24. 2500×400
.....

25. 1750×40
.....

26. 4500×200
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Estimate, Then Compute

Round both, multiply, then do it properly and compare.

Warm-Up 12 ITEMS • RECALL

1 Estimate 19×21

2 Estimate 48×52

3 Estimate 31×29

4 Estimate 62×18

5 Estimate 78×22

6 Estimate 39×11

7 Estimate 29×31

8 Estimate 41×62

9 Estimate 18×22

10 Estimate 77×33

11 Estimate 51×48

12 Estimate 91×12

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Estimate 412×19

.....

14. True value of 412×19

.....

15. Estimate 289×31

.....

16. True value of 289×31

.....

17. Estimate 197×203

.....

18. Estimate 508×21

.....

19. True value of 508×21

.....

20. Estimate 392×41

.....

21. True value of 392×41

.....

22. Estimate 297×103

.....



Working space



Error journal

Four Digits by Two

Longer, not harder. Keep the estimate in your head.

Warm-Up 12 ITEMS • RECALL

1 1000×10

2 2000×20

3 1200×10

4 3000×30

5 1500×20

6 2500×10

7 2000×10

8 1500×10

9 3000×20

10 4000×30

11 2500×20

12 1200×30

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 1234×12

14. 2450×36

15. 1875×24

16. 3081×47

17. 4206×15

18. 2143×13

19. 3260×24

20. 1450×32

21. 5012×26

22. 3105×18

★ Challenge NOT GRADED • REASON IT OUT

23. 5678×43

.....

24. 9999×99

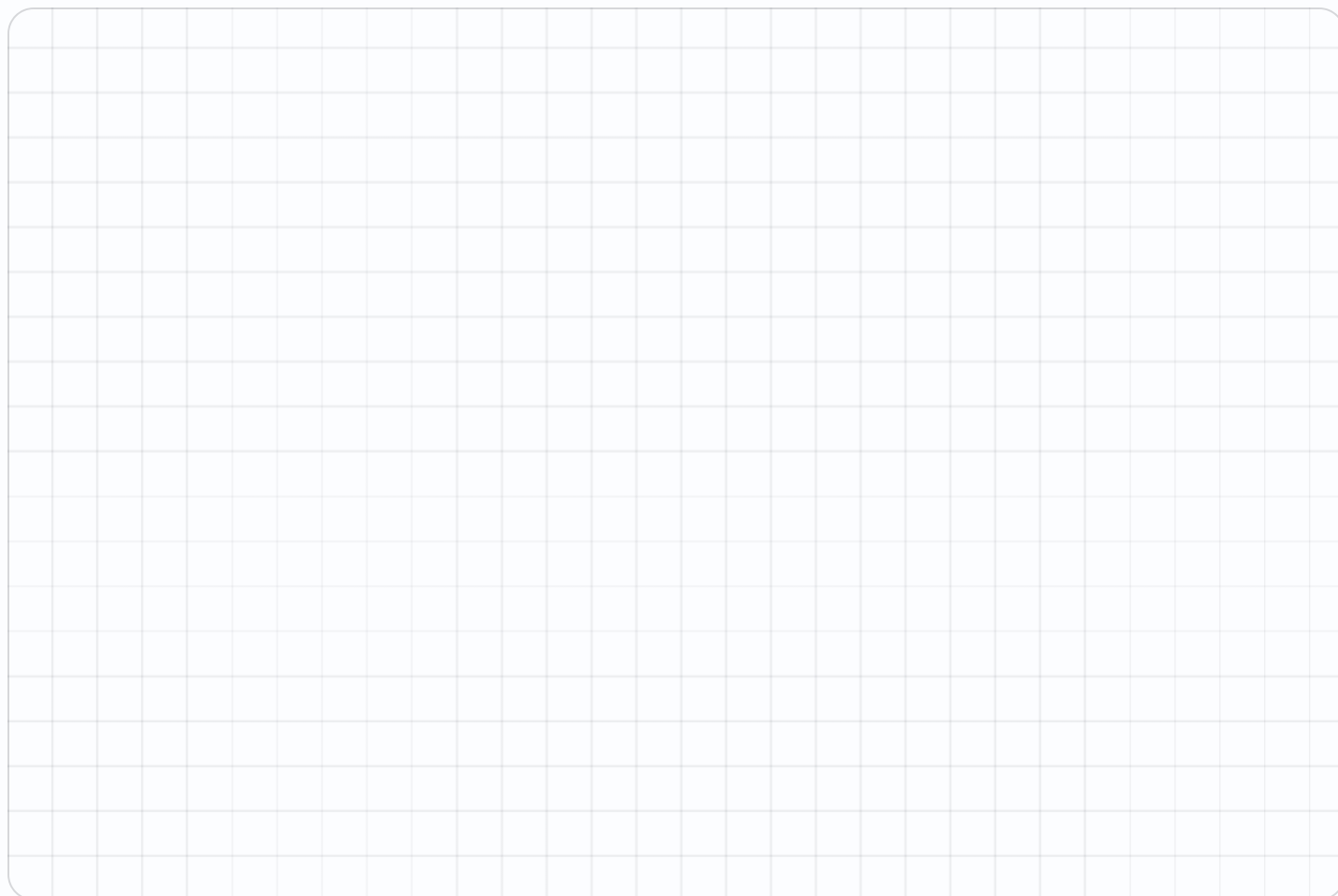
.....

25. 6789×34

.....

26. 8888×88

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Catch the Error

Given a wrong worked answer, find the line it went wrong.

Warm-Up 12 ITEMS • RECALL

1 23×14

2 23×4

3 23×10

4 $92 + 230$

5 31×22

6 45×23

7 34×12

8 34×2

9 34×10

10 $68 + 340$

11 41×21

12 52×13

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Someone got $23 \times 14 = 92$. What did they forget — type the missing partial

.....

14. Someone got $45 \times 23 = 135$. The real answer

.....

15. Someone got $38 \times 24 = 152$. The real answer

.....

16. Someone got $56 \times 27 = 392$. The real answer

.....

17. 63×48

.....

18. Someone got $34 \times 12 = 68$. The missing partial

.....

19. Someone got $52 \times 13 = 156$. The real answer

.....

20. Someone got $47 \times 26 = 282$. The real answer

.....

21. Someone got $63 \times 35 = 315$. The real answer

.....

22. 71×44

.....

Mid-Unit Quiz

Eight items across Weeks 1–2. 85% to keep flying.

✦ Warm-Up 4 ITEMS • RECALL

1 40×60

2 23×14

3 50×40

4 34×12

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. 45×23

6. 237×45

7. Estimate 412×19

8. 1234×12

9. 506×34

10. 52×13

11. 318×52

12. Estimate 508×21

13. 2143×13

14. 426×23

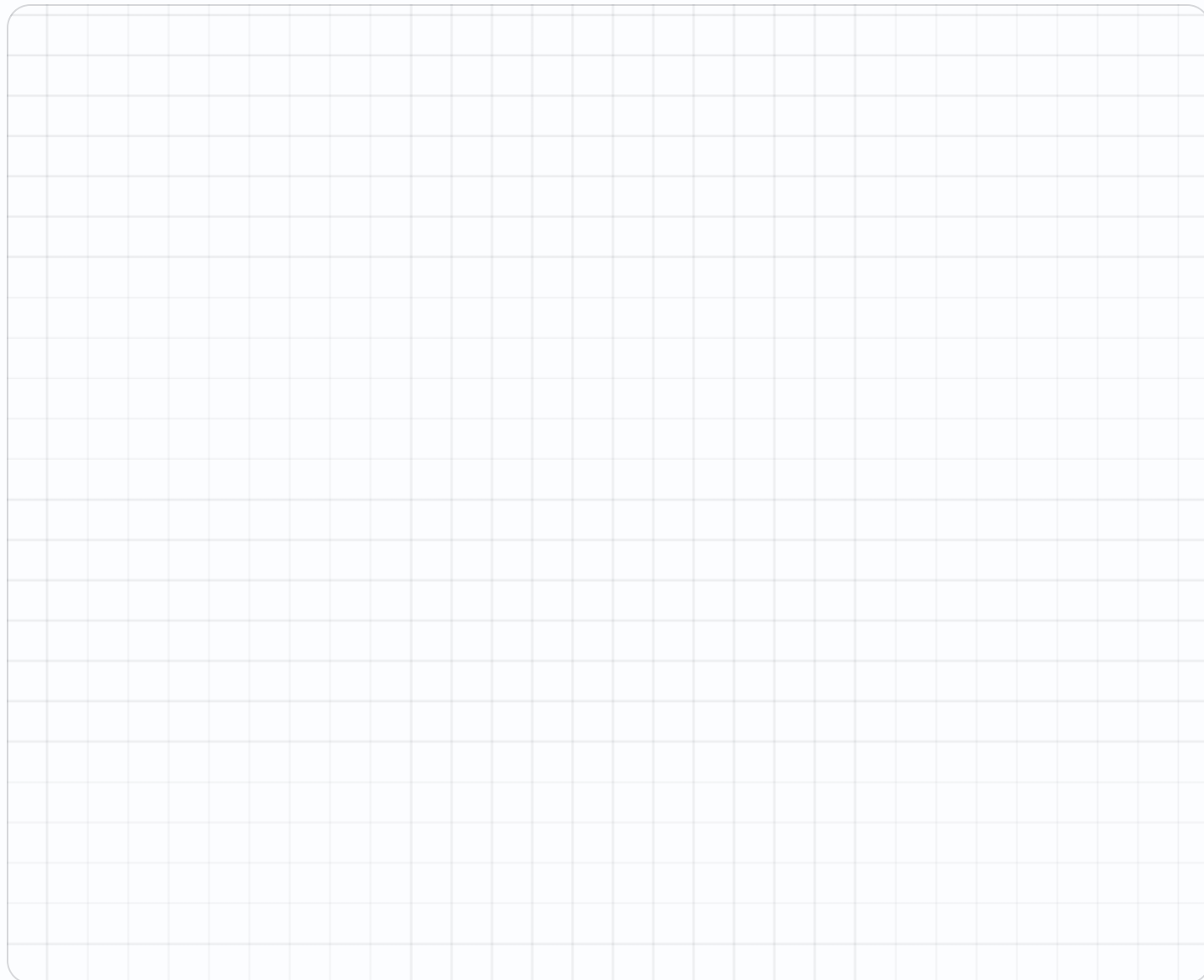
★ Challenge NOT GRADED • REASON IT OUT

15. 2450×36

.....

16. 3260×24

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Mark the reasoning, not the handwriting.

Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

2.1 • Multiples of Ten First

OUT OF 22

Warm-Up • 1-12 • 600 | 2400 | 2500 | 2100 | 3200 | 8100 | 900 | 1000 | 2400 | 5600 | 1800 | 3000

Core • 13-22 • 28000 | 300000 | 10000 | 7200 | 7000 | 24000 | 210000 | 7000 | 7000 | 10000

Challenge • 23-26 • 100000 | 1000000 | 70000 | 900000

2.2 • Estimate, Then Compute

OUT OF 22

Warm-Up • 1-12 • 400 | 2500 | 900 | 1200 | 1600 | 400 | 900 | 2400 | 400 | 2400 | 2500 | 900

Core • 13-22 • 8000 | 7828 | 9000 | 8959 | 40000 | 10000 | 10668 | 16000 | 16072 | 30000

Challenge • 23-26 • 250000 | 884 | 240000 | 1612

2.3 • Four Digits by Two

OUT OF 22

Warm-Up • 1-12 • 10000 | 40000 | 12000 | 90000 | 30000 | 25000 | 20000 | 15000 | 60000 | 120000 | 50000 | 36000

Core • 13-22 • 14808 | 88200 | 45000 | 144807 | 63090 | 27859 | 78240 | 46400 | 130312 | 55890

Challenge • 23-26 • 244154 | 989901 | 230826 | 782144

2.4 • Catch the Error

OUT OF 22

Warm-Up • 1-12 • 322 | 92 | 230 | 322 | 682 | 1035 | 408 | 68 | 340 | 408 | 861 | 676

Core • 13-22 • 230 | 1035 | 912 | 1512 | 3024 | 340 | 676 | 1222 | 2205 | 3124

Challenge • 23-26 • 10665 | 9480 | 16536 | 15900

Fri • Mid-Unit Quiz

OUT OF 14

Warm-Up • 1-4 • 2400 | 322 | 2000 | 408

Core • 5-14 • 1035 | 10665 | 8000 | 14808 | 17204 | 676 | 16536 | 10000 | 27859 | 9798

Challenge • 15-16 • 88200 | 78240

Round to Estimate

Which way to round, and what it costs you.

Warm-Up 12 ITEMS • RECALL

1 Estimate 19×21

2 Estimate 48×52

3 Estimate 31×29

4 Estimate 62×18

5 Estimate 78×22

6 Estimate 39×11

7 Round 47 to the nearest ten

8 Round 312 to the nearest hundred

9 Round 89 to the nearest ten

10 Round 651 to the nearest hundred

11 Round 2,400 to the nearest thousand

12 Round 55 to the nearest ten

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Estimate 412×19

.....

14. True value of 412×19

.....

15. Estimate 289×31

.....

16. True value of 289×31

.....

17. Estimate 197×203

.....

18. Estimate 47×31 by rounding both

.....

19. True value of 47×31

.....

20. Estimate 312×89

.....

21. True value of 312×89

.....

22. Estimate 651×48

.....



Working space



Error journal

Over or Under

Say before you compute whether your estimate is high or low.

Warm-Up 12 ITEMS • RECALL

1 Rounding both numbers up gives an estimate that is — over or under

2 Rounding both down gives — over or under

3 19×21

4 48×52

5 20×30

6 40×60

7 Rounding 47 up to 50 makes the estimate — type over or under

8 Rounding 42 down to 40 makes it — type over or under

9 Estimate 48×52 by rounding both to 50

10 True value of 48×52

11 Estimate 41×39

12 True value of 41×39

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Estimate 412×19 as 400×20

14. True value

15. Was the estimate over or under

16. Estimate 78×22 as 80×20

17. True value of 78×22

18. Rounding both numbers up makes the estimate — type over or under

19. Estimate 58×63 rounding both up

20. True value of 58×63

21. Estimate 29×31 rounding both to 30

22. True value of 29×31

★ Challenge NOT GRADED • REASON IT OUT

23. 63×48

24. Estimating that as $60 \times 50 = 3000$ — over or under

25. Why does rounding one up and one down land close — type cancel or add

26. Estimate 199×201 and give the true value

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal ONE SENTENCE PER MISTAKE

Order of Magnitude

Is it hundreds, thousands or tens of thousands?

Warm-Up 12 ITEMS • RECALL

1 30×40

2 300×40

3 300×400

4 3×4

5 30×4

6 3000×4

7 Zeros in 100
_____8 Zeros in 1,000
_____9 20×30 — zeros in the answer

10 200×300

11 Zeros in 20×500

12 40×25

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Estimate 34×26 to the nearest hundred
.....14. Estimate 340×26
.....15. Estimate $3,400 \times 26$
.....16. Digits in the answer to 34×26
.....17. Digits in the answer to 340×26
.....18. About how big is 40×60 — type hundreds or thousands
.....19. About how big is 400×600 — type thousands or hundred thousands
.....20. Digits in the answer to 300×400
.....21. Digits in the answer to 50×50
.....22. Someone gets 812 for 40×60 . Digits it should have
.....



Working space



Error journal

Mark Somebody's Work

Six worked answers, three wrong. Find them by estimating.

Warm-Up 11 ITEMS • RECALL

1 23×14

2 45×23

3 31×22

4 38×24

5 56×27

6 12×12

7 25×4

8 15×6

9 32×3

10 20×14

11 11×13

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. 23×14 given as 92 — the missing partial product
.....13. 45×23 given as 135 — the missing partial product
.....14. 237×45 given as 1,185 — the missing partial product
.....15. 56×27 with the second row as 112 — it should be
.....16. That answer is out by
.....17. Someone got $25 \times 14 = 100$. The real answer
.....18. Someone got $43 \times 21 = 129$. The real answer
.....19. Someone got $36 \times 25 = 180$. The real answer
.....20. Someone got $54 \times 32 = 1,068$. The real answer
.....21. 84×26
.....



Challenge

22. 78×46

23. 94×68

24. Someone got $209 \times 43 = 627$. The missing partial

25. 209×43



Working space



Error journal

Stadium Count Begins

Estimate from the photo before looking anything up.

Warm-Up 8 ITEMS • RECALL

1 24 seats $\times$ 38 rows2 20×25 3 500×16 4 30×30 5 40×25 6 1000×16

7 A section of 20 rows of 30 seats

8 Four such sections

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

9. 38×24 10. 912×16 11. Estimate $40 \times 25 \times 16$

12. The gap between that estimate and the true count

13. $38 \times 24 \times 8$

14. A section of 42 rows of 38 seats

15. Six such sections

Challenge NOT GRADED • REASON IT OUT

16. $2,450 \times 36$

17. $1,875 \times 24$

18. A stadium of 24 sections of 1,850 seats

19. That stadium 80% full — seats taken

20. Estimate $24 \times 1,850$ by rounding

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 3 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

3.1 • Round to Estimate

OUT OF 22

Warm-Up • 1-12 • 400 | 2500 | 900 | 1200 | 1600 | 400 | 50 | 300 | 90 | 700 | 2000
| 60

Core • 13-22 • 8000 | 7828 | 9000 | 8959 | 40000 | 1500 | 1457 | 27000 | 27768 |
35000

Challenge • 23-26 • 250000 | 90000 | 160000 | 190548

3.2 • Over or Under

OUT OF 22

Warm-Up • 1-12 • over | under | 399 | 2496 | 600 | 2400 | over | under | 2500 | 2496
| 1600 | 1599

Core • 13-22 • 8000 | 7828 | over | 1600 | 1716 | over | 3600 | 3654 | 900 | 899

Challenge • 23-26 • 3024 | under | cancel | 39999

3.3 • Order of Magnitude

OUT OF 22

Warm-Up • 1-12 • 1200 | 12000 | 120000 | 12 | 120 | 12000 | 2 | 3 | 2 | 60000 | 4
| 1000

Core • 13-22 • 900 | 9000 | 90000 | 3 | 4 | thousands | hundred thousands | 6 | 4 | 4

Challenge • 23-26 • 884 | 10665 | 8 | 10000000

3.4 • Mark Somebody's Work

OUT OF 21

Warm-Up • 1-11 • 322 | 1035 | 682 | 912 | 1512 | 144 | 100 | 90 | 96 | 280 | 143

Core • 12-21 • 230 | 900 | 9480 | 1120 | 1008 | 350 | 903 | 900 | 1728 | 2184

Challenge • 22-25 • 3588 | 6392 | 8360 | 8987

Fri • Stadium Count Begins

OUT OF 15

Warm-Up • 1-8 • 912 | 500 | 8000 | 900 | 1000 | 16000 | 600 | 2400

Core • 9-15 • 912 | 14592 | 16000 | 1408 | 7296 | 1596 | 9576

Challenge • 16-20 • 88200 | 45000 | 44400 | 35520 | 40000

Count a Section

Seats per row, rows per section, sections per stadium.

Warm-Up 12 ITEMS • RECALL

1 24×2

2 24×10

3 24×20

4 24×30

5 24×38

6 12×24

7 15 rows of 20

8 25 rows of 20

9 18 rows of 10

10 30 rows of 30

11 12 rows of 25

12 40 rows of 15

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 26×42

14. 32×48

15. 28×36

16. 30×45

17. 22×54

18. 27 rows of 34

19. 46 rows of 28

20. 53 rows of 41

21. 38 rows of 45

22. 64 rows of 36

Challenge

23. $38 \times 24 \times 4$

24. $38 \times 24 \times 16$

25. A block of 125 rows of 48

26. That block minus 220 broken seats



Working space



Error journal

Scale It Up

One section times the number of sections.

Warm-Up 12 ITEMS • RECALL

1 900×10

2 900×16

3 1000×16

4 500×16

5 900×2

6 900×4

7 One section of 500, times 4

8 Times 10

9 600×8

10 250×4

11 1000×12

12 750×2

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 912×16

14. 912×8

15. $1,092 \times 12$

16. $1,536 \times 10$

17. $1,008 \times 20$

18. $1,850 \times 12$

19. $2,400 \times 18$

20. $3,150 \times 22$

21. $1,275 \times 16$

22. 960×45



Working space



Error journal

Explain the Gap

Compare to the published capacity and account for the difference.

Warm-Up 12 ITEMS • RECALL

1 16,000 – 14,592

2 900 – 884

3 15,000 – 14,592

4 16,000 – 15,000

5 1000 – 912

6 2500 – 2496

7 Estimated 2,000, actual 2,150 — the gap

8 Estimated 500, actual 480 — the gap

9 Estimated 1,000, actual 1,120 — the gap

10 6,000 – 5,850

11 Estimated 300, actual 300 — the gap

12 12,000 – 11,600

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Estimating 38×24 as 40×25 — over by

.....

14. That gap multiplied by 16

.....

15. 1,408 as a percent of 14,592, to the nearest whole

.....

16. Estimating 197×48 as 200×50 — the estimate

.....

17. True value of 197×48

.....

18. Estimated 40,000, actual 44,400 — the gap

.....

19. That gap as a percent of 44,400, to the nearest whole

.....

20. Estimated 27,000, actual 27,768 — the gap

.....

21. Rounding both numbers up gives an estimate that is — type high or low

.....

22. Estimated 9,000, actual 8,959 — the gap

.....

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

✦ Warm-Up 10 ITEMS • RECALL

1 7×8

2 40×60

3 23×14

4 12×6

5 400×70

6 9×9

7 25×4

8 12×12

9 300×20

10 50×50

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. 45×23

12. 237×45

13. 506×34

14. $1,234 \times 12$

15. Estimate 412×19

16. 34×27

17. 52×19

18. 1234×12

19. Estimate 489×29

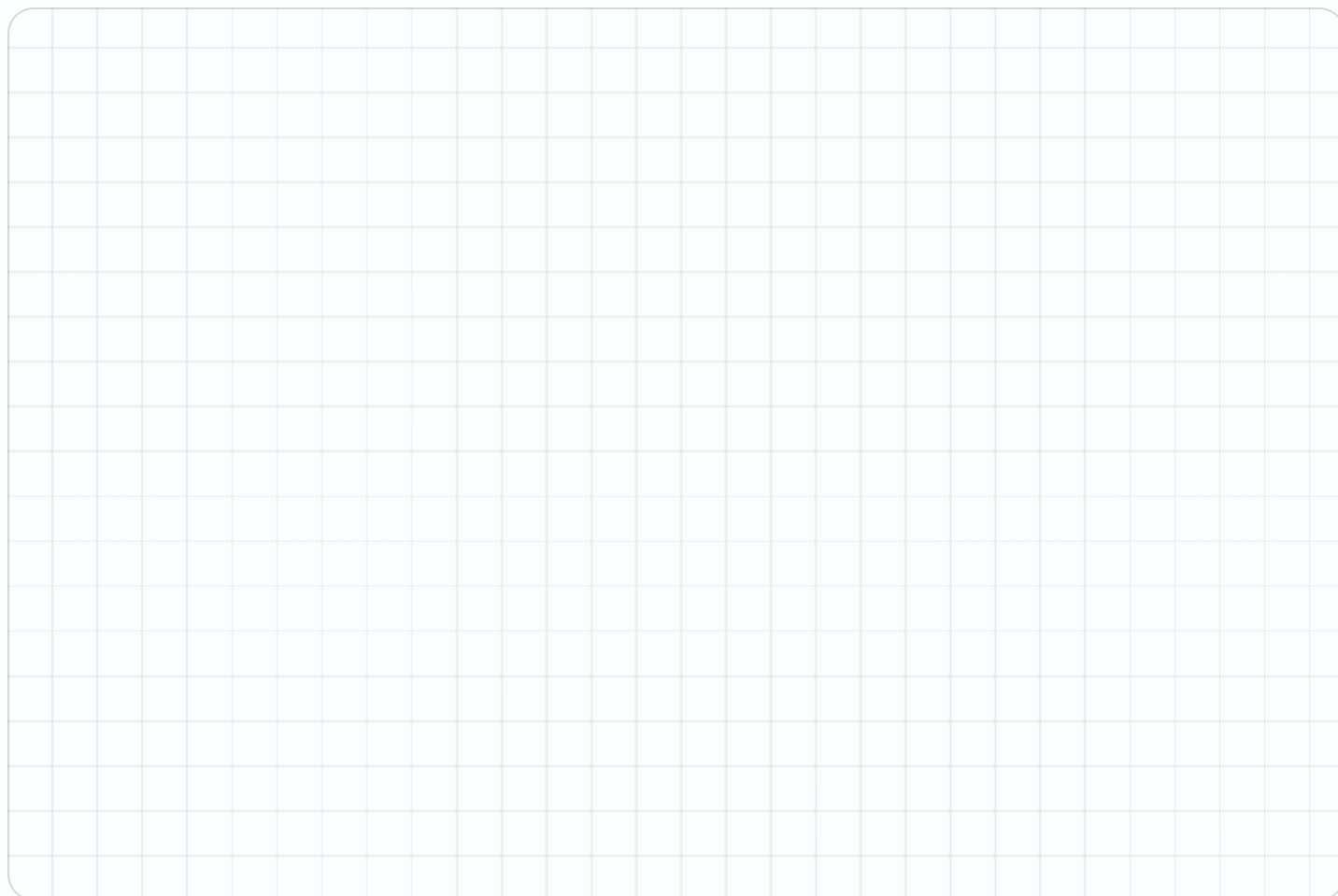
★ Challenge NOT GRADED • REASON IT OUT

20. $9,999 \times 99$
.....

21. $2,450 \times 36$
.....

22. 2450×36
.....

23. 5678×43
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Mission 02 Test

Twelve items plus the Big Question, answered out loud.

Warm-Up 4 ITEMS • RECALL

1 50×50

2 38×24

3 30×40

4 21×12

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. 56×27

6. 289×31

7. 125×24

8. $2,450 \times 36$

9. Estimate $5,120 \times 48$

10. Estimate 197×203

11. $1,875 \times 24$

12. $9,999 \times 99$

13. 47×23

14. 312×21

15. Estimate 58×63

16. 2143×13

17. Someone got $52 \times 13 = 156$. The real answer

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • Count a Section

OUT OF 22

Warm-Up • 1-12 • 48 | 240 | 480 | 720 | 912 | 288 | 300 | 500 | 180 | 900 | 300 | 600

Core • 13-22 • 1092 | 1536 | 1008 | 1350 | 1188 | 918 | 1288 | 2173 | 1710 | 2304

Challenge • 23-26 • 3648 | 14592 | 6000 | 5780

4.2 • Scale It Up

OUT OF 22

Warm-Up • 1-12 • 9000 | 14400 | 16000 | 8000 | 1800 | 3600 | 2000 | 5000 | 4800 | 1000 | 12000 | 1500

Core • 13-22 • 14592 | 7296 | 13104 | 15360 | 20160 | 22200 | 43200 | 69300 | 20400 | 43200

Challenge • 23-26 • 15000 | 18240 | 116400 | 57800

4.3 • Explain the Gap

OUT OF 22

Warm-Up • 1-12 • 1408 | 16 | 408 | 1000 | 88 | 4 | 150 | 20 | 120 | 150 | 0 | 400

Core • 13-22 • 88 | 1408 | 10 | 10000 | 9456 | 4400 | 10 | 768 | high | 41

Challenge • 23-26 • 39991 | 9 | yes | hundred

Thu • Error Journal Sweep

OUT OF 19

Warm-Up • 1-10 • 56 | 2400 | 322 | 72 | 28000 | 81 | 100 | 144 | 6000 | 2500

Core • 11-19 • 1035 | 10665 | 17204 | 14808 | 8000 | 918 | 988 | 14808 | 15000

Challenge • 20-23 • 989901 | 88200 | 88200 | 244154

Fri • Mission 02 Test

OUT OF 20

Warm-Up • 1-4 • 2500 | 912 | 1200 | 252

Core • 5-20 • 1512 | 8959 | 3000 | 88200 | 250000 | 40000 | 45000 | 989901 | 1081 | 6552 | 3600 | 27859 | 676 | 22200 | 2304 | 9798

Challenge • 21-24 • 9480 | 14592 | 230826 | 57800